Purely Magnetically Driven Flow

The chosen field, the transverse force balance, and the solver architecture

MTP — Negative rotational viscosity in ferrofluids via phase-field MRT-LBM

The goal of this phase

Until now the flow was pushed by a body force $\mathbf{F}_b$ standing in for a pressure gradient. Here we **delete the body force** ($\mathbf{F}_{b,x} = 0$) and let the *magnetic Kelvin force* do all the pumping, from a spatially graded applied field. The oscillating part can then modulate the viscosity on top. The obvious worry — "a field with a sideways (y) component should push fluid sideways too" — turns out to be answered cleanly: the sideways force is quietly cancelled by pressure, and only the down-channel force survives to drive flow. This PDF derives the field, proves the balance, and lays out the full solver.

1 The magnetic field we chose, and why

Without fluid ($\mathbf{M} = 0$) the applied potential must be harmonic, $\nabla^2 A_{\text{app}} = 0$. Expanding to the lowest order that gives a *non-uniform* field, in channel-centred coordinates $X = x - x_0$, $Y = y - y_0$:

$$A_{\text{app}} = \mu_0 \Big[\alpha X + \beta Y + \tfrac{1}{2}(aX^2 + 2bXY + cY^2) \Big]. \tag{1}$$

Two physical requirements fix the coefficients:

- **Harmonic** ($\nabla^2 A_{\text{app}} = 0$): $a + c = 0$.
- **Symmetric about the centreline** (under $Y \to -Y$, H_x even and H_y odd): $c = 0$, $\alpha = 0$, hence $a = 0$.

Renaming $\beta = H_d$ (uniform part) and $b = k$ (gradient), the field is

$$\boxed{A_{\text{app}} = \mu_0(H_d\, Y + k\, XY), \qquad \mathbf{H}_{\text{app}} = \big(H_d + kX,\ -kY\big)}. \tag{2}$$

Why H_d cannot be zero

It is tempting to set the uniform part $H_d = 0$ and drive purely with the gradient. But when $\mathbf{M} \parallel \mathbf{H}$ and $\nabla \times \mathbf{H} = 0$, the Kelvin force is $\mathbf{F}_K = \mu_0 M\, \nabla|\mathbf{H}|$ — it pushes fluid toward *stronger* field. With $H_d = 0$, $|\mathbf{H}| = k\sqrt{X^2 + Y^2}$ has a **null at the channel centre** and grows in *every* direction outward. The force then pushes fluid *away from the centre both ways*, and the net down-channel drive is zero. So we need $H_d > kL/2$ (no null inside) and $H_d \gg kL$ (for the seam condition). Then $|\mathbf{H}| \approx H_d + kX$ increases along x and the force points cleanly down the channel.

With $H_d \gg kL$:

$$F_{K,x} \approx \mu_0 M(H_d)\, k \quad \text{(nearly uniform, replaces } \mathbf{F}_b\text{)}, \qquad \frac{F_{K,y}}{F_{K,x}} \approx \frac{kY}{H_d} \ll 1. \tag{3}$$

At rest $\mathbf{M} \times \mathbf{H} = 0$ so the rotational force is off; once the fluid flows, vorticity tilts $\mathbf{M}$ and it switches on — the mechanism that lets the oscillating field modulate viscosity.

1.1 Representative values

Fixing $H_d = 1.73$, $\xi_d = 20$ (so $M_s = 0.01734$, $M(H_d) = 0.0165$) and matching the old body force $G = 2 \times 10^{-7}$ gives

$$k = \frac{G}{\mu_0 M(H_d)} = 1.21 \times 10^{-5}, \quad \frac{kL}{H_d} = 3.5 \times 10^{-4}, \quad \frac{F_{K,y}}{F_{K,x}} \le \frac{kH}{2H_d} = 1.8 \times 10^{-4}. \tag{4}$$

2 Why the flow stays a clean parabola even though $F_y \neq 0$

The bucket of water

Water in a glass has gravity pulling every drop straight down — hard. Yet the water does not flow downward. Why? A vertical pressure gradient (higher at the bottom) exactly balances gravity. **The force is there; the flow is not, because the force is balanced.** Our sideways magnetic force F_y is the "gravity"; the fluid builds a sideways pressure gradient that is the "water's response," and the net sideways motion is zero.

2.1 The momentum equations decouple

For steady, fully developed, unidirectional flow $\mathbf{u} = (u(y), 0)$, continuity holds identically and Navier–Stokes separates:

$$x: \quad 0 = -\partial_x p + \eta\, u''(y) + F_x, \tag{5}$$

$$y: \quad 0 = -\partial_y p + F_y(y). \tag{6}$$

Look at the y-equation: no u, no u', no u''. It is *not a statement about motion* — it is a balance between two sideways pushes. Whatever $F_y(y)$ is, pressure arranges itself to cancel it:

$$\partial_y p = F_y(y) \implies p(x,y) = -Gx + \int^y F_y(y')\, dy' + \text{const}.$$

The fluid is slightly compressed toward one wall, and that transverse pressure holds F_y in check. Net sideways flow: **zero**.

2.2 Why x is different — periodicity blocks the same trick

In x the channel is periodic, $p(x + L, y) = p(x, y)$, so pressure has *no net slope* along x. Split $F_x = \bar{F}_x + F_x'(x)$: pressure can absorb the periodic wiggle $F_x'(x)$ but *not* the mean $\bar{F}_x$. The leftover mean is the only thing that drives flow:

$$\eta\, u''(y) = -\bar{F}_x \implies u(y) = \frac{\bar{F}_x}{2\eta}\, y(H - y) \quad \text{(the parabola)}. \tag{7}$$

The deep reason: curl

Pressure is a gradient ∇p, and gradients have zero curl. So **a force drives flow only if it has curl that survives averaging**. $F_y(y)$ depends on y *alone* — a single-coordinate force is a pure gradient, expressible as ∇(something), so pressure swallows it whole. $\bar{F}_x$ survives because periodic geometry forbids pressure from developing the mean slope needed to cancel it. That is the real asymmetry between x and y.

2.3 And here F_y is doubly harmless

Working out the transverse Kelvin force with the forced $H_y = -k(y - y_c)$ and aligned M_y:

$$F_{K,y} = \mu_0(M_x \partial_x H_y + M_y \partial_y H_y) = -\mu_0 k M_y \approx \mu_0 k^2 M \, \frac{y - y_c}{H_d} = O(k^2). \tag{8}$$

It is (i) **second order in the gradient** — smaller than the first-order drive $\bar{F}_x = \mu_0 k \langle M_x \rangle$ by a factor $kH/H_d \sim 10^{-4}$; and (ii) a **function of y only** — hence a pure gradient, cancelled exactly by pressure.

The transverse velocity measured $|u_y| \leq 1.5 \times 10^{-9}$ — essentially machine noise — against $u_{\max} \sim 10^{-3}$. The parabola is undisturbed.

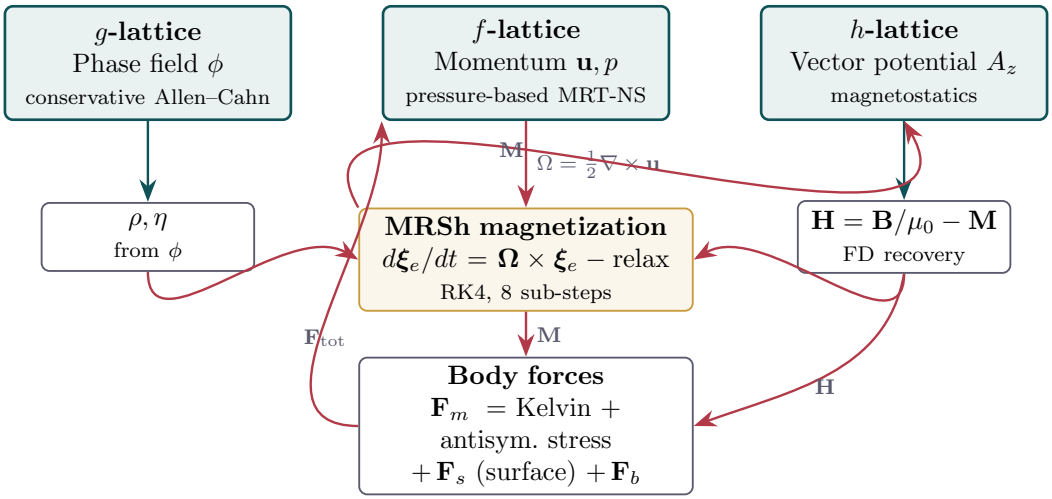
The clean decoupling needs the flow to be unidirectional and fully developed, $\mathbf{u} = (u(y), 0)$. If F_y varied with x as well as y, its curl would not vanish, pressure could not fully absorb it, and secondary transverse cells would appear. It does not here, because $\mathbf{H}_{\text{app}}$ is built to keep F_y a function of y alone.

2.4 The transverse-field antisymmetric stress

There is a genuine extra body force from the transverse field forced by $\nabla \cdot \mathbf{H} = 0$: an antisymmetric-stress contribution εG that scales *linearly* with k and is *independent of the drive amplitude* (confirmed by an amplitude sweep). It shifts the effective drive but does not break the parabolic profile.

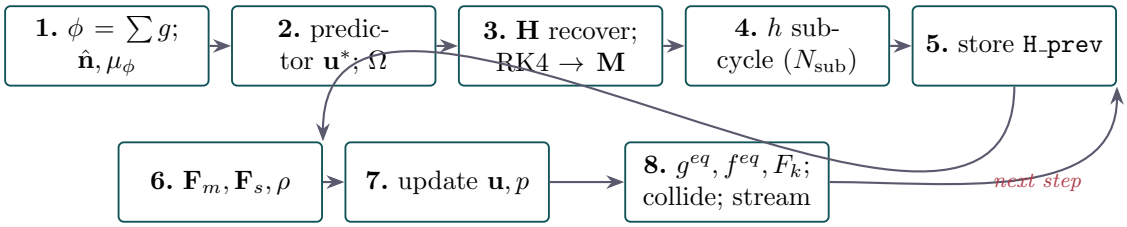
3 Architecture of the LBM simulation

THREE COUPLED LATTICES



The loop closes: flow → vorticity tilts magnetization → magnetization sets force → force drives flow.

ONE TIMESTEP (order of operations)



- Applied field $\mathbf{H}_{\text{app}} = (H_d + kX, -kY)$ from $A_{\text{app}} = \mu_0(H_d Y + kXY)$; harmonic and centreline-symmetric by construction.
- $H_d \neq 0$ is mandatory — a pure gradient field has a null at the centre and produces zero net drive.
- Kelvin force $F_{K,x} \approx \mu_0 M(H_d)k$ replaces the body force; $F_{K,y} = O(k^2)$, a y-only pure gradient, cancelled exactly by transverse pressure ($|u_y| \lesssim 10^{-9}$).
- The seam ($x = 0, L$) uses a jump-periodic condition on A_z, since A_{app} is not periodic; the induced part is periodic when $H_d \gg kL$.
- Three lattices (g, f, h) coupled through MRSh magnetization and the Kelvin+rotational force close the loop each timestep.